

Generative AI–Powered Childcare Affordability Assistant: Refined Design and Prompt

Experiments

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DSC 670: Advanced Uses of Generative AI

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July 2026

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Childcare costs create a serious financial burden for many families, but a generative artificial intelligence (AI) system cannot directly change tuition rates or make providers charge less. Based on the feedback from Milestone 1, I refined the project to focus on an information and decision-support problem. Families may have difficulty finding, comparing, and understanding lower-cost childcare options, subsidy programs, and the effect of childcare expenses on a household budget. The proposed Childcare Affordability Assistant is intended to reduce the effort required to interpret this information. It would not promise to make childcare cheaper. Instead, it would help a user identify potentially lower-cost options and financial assistance that may reduce the user's individual childcare burden.

The refined project uses a large language model (LLM) as the conversational component and Retrieval-Augmented Generation (RAG) as the preferred architecture for grounding responses in childcare information. The five prompt experiments test whether an LLM can explain costs, compare options, discuss assistance, acknowledge missing information, and provide structured decision support. I evaluated the responses for accuracy, relevance, completeness, consistency, and whether the model stayed within the intended scope.

Refined Problem Statement

The problem is not that families lack all childcare information. Cost reports, provider resources, and public assistance information already exist. The problem is that the information is distributed across multiple sources and can be difficult to compare or interpret. Based on the Milestone 1

feedback, I refined the project so it does not claim that generative AI can lower childcare prices. Families instead need a clearer and more efficient way to find, compare, and understand childcare cost information and potential financial assistance.

The proposed assistant will summarize relevant information, ask for missing context, compare options, and explain possible next steps. It will not claim to lower market childcare prices or guarantee eligibility for assistance. Its purpose is to improve access to information so users can identify potentially lower-cost choices and programs that may reduce their individual childcare burden.

Selected Generative AI Approach

I plan to use a large language model (LLM) for natural-language interaction and Retrieval-Augmented Generation (RAG) to supply relevant childcare documents and data at response time. An LLM is useful because users ask the same type of question in many different ways. RAG is important because childcare costs and assistance rules vary by location and can change over time.

In a RAG workflow, the system retrieves relevant information from an approved knowledge base and provides that context to the language model before the answer is generated. The knowledge base could include government childcare assistance information, state or county program documents, childcare cost reports, and selected nonprofit resources. I prefer RAG over using fine-tuning to memorize current prices and program rules because the source collection can be

updated as information changes. Fine-tuning may still be useful later for response style and consistent decision-support behavior.

Prompt Experiments

Experiment 1: Zero-Shot Cost Explanation

Prompt: A parent asks: 'Why does childcare feel unaffordable even when I have a full-time job?' Explain the issue in plain language. Do not claim that the AI can lower childcare prices. Explain how a childcare decision-support assistant could help.

The response should distinguish the actual price of childcare from the information problem. I selected this prompt to test the refined scope of the project. A successful response should explain affordability concerns without suggesting that the AI changes provider prices. This experiment is intentionally broad, so I expect a useful explanation but not a personalized recommendation.

Experiment 2: Personalized Affordability Comparison

Prompt: A family earns \$5,000 per month after taxes. They are comparing center-based infant care at \$1,600 per month and family childcare at \$1,150 per month. Compare the two options as percentages of monthly income, explain the monthly difference, and discuss affordability without making the decision for the family.

The expected calculations are 32% of monthly income for center-based care and 23% for family childcare, with a \$450 monthly difference. I chose this experiment because numerical results can be manually verified. I also instructed the model not to make the final decision because cost is only one factor in a childcare choice.

Experiment 3: Financial Assistance Discovery

Prompt: A parent says: 'My childcare bill is too high and I do not know where to start looking for help.' Ask for only the information needed to narrow the search for childcare assistance. Do not promise eligibility. Then explain the types of official programs or resources that may be checked. This prompt tests whether the model recognizes that assistance recommendations require context. I expect it to ask for location and basic household information before narrowing the search. A generic list may be less useful, and the model should not promise eligibility without current program evidence.

Experiment 4: Missing-Information and Uncertainty Test

Prompt: A user asks: 'How much is infant childcare in my area and which subsidy will I qualify for?' The user has not provided a city, state, ZIP code, income, or household size. Respond safely and helpfully. Do not invent local prices or determine eligibility without evidence.

This experiment tests whether the model knows when information is missing. I consider this especially important because a fluent answer can still be wrong. A successful result should identify the information gap, explain the limitation, and ask focused follow-up questions instead of inventing a local cost or subsidy determination.

Experiment 5: Structured Decision Support

Prompt: Compare three childcare options for a family: Center A costs \$1,500 per month and is 10 minutes away; Family Provider B costs \$1,100 per month and is 25 minutes away; Center C costs \$1,300 per month and is 15 minutes away. Create a concise comparison using cost, commute, and information still needed. Identify the lowest-cost option, but do not call it the 'best' option unless enough quality and scheduling information is available.

This prompt tests a multi-factor comparison. Family Provider B is clearly the lowest-cost option, but price alone does not establish that it is the best choice. I expect the model to identify missing factors such as quality, hours, availability, licensing information, and the child's needs.

Comparison of Prompt Experiment Results

The five experiments are similar because each uses natural-language instructions to define the model's task and expected behavior. The main differences are the amount of context and the level of constraint. Experiment 1 is broad and explanatory. Experiment 2 introduces arithmetic. Experiment 3 requires the model to ask for missing context before narrowing assistance resources. Experiment 4 specifically tests uncertainty and refusal to invent facts. Experiment 5 requires a multi-factor comparison and places a constraint on the word “best.”

The experiments suggest that specific prompts produce responses that are easier to evaluate. Experiments 2 and 5 contain facts that can be manually checked, while Experiments 3 and 4 test whether the model recognizes missing information. The strongest prompts clearly define the task, state what the model should not claim, and describe the expected response. A successful prompt demonstration is still not a complete childcare application. Current local costs and assistance rules require reliable external data, which supports my decision to use RAG.

Proposed Fine-Tuning and Customization Strategy

I would separate factual grounding from behavioral customization. I would use RAG for current childcare costs, program rules, and location-specific information. I would consider supervised fine-tuning only after collecting a high-quality set of reviewed example conversations that

demonstrate the behavior expected from the assistant.

Training examples would teach the model to ask for location when local cost information is requested, avoid guaranteeing subsidy eligibility, distinguish “lowest cost” from “best,” explain affordability calculations in plain language, and state when retrieved evidence is insufficient.

Before fine-tuning, I would establish a baseline using prompt engineering and RAG. I would compare the baseline and customized model on factual accuracy, calculation accuracy, relevance, completeness, consistency, and appropriate uncertainty. Human review would remain necessary for assistance eligibility and major financial decisions.

Limitations and Safeguards

Generative AI can produce confident language when information is incomplete. Childcare prices, provider availability, and assistance requirements can change. The system should retrieve dated information from approved sources, distinguish estimates from confirmed facts, and encourage users to verify eligibility or enrollment details with the responsible agency or provider.

Affordability is also personal. A lower monthly price does not automatically make a childcare option suitable. Commute time, hours, quality, licensing, availability, and family preferences may change the decision. The assistant should present comparisons and explain tradeoffs rather than making the final decision for the user.

Conclusion

Milestone 2 refines the Childcare Affordability Assistant from a broad idea about making childcare more affordable into a realistic generative AI decision-support project. The assistant cannot reduce provider prices. Its value is helping families find, compare, and understand childcare cost information and potential assistance resources. The five prompt experiments test explanation, arithmetic, missing-context recognition, uncertainty, and multi-factor comparison.

The preferred design combines an LLM with RAG so responses can be grounded in approved and updateable childcare information. Fine-tuning may later improve behavioral consistency, but it should not replace retrieval for changing factual information. The next phase should focus on building an approved knowledge base and evaluating generated responses against known source documents and calculations.

References

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